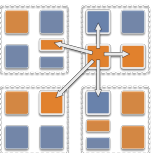


Exercise 2

Goal: implement k-means clustering using the classic Lloyd iterative refinement scheme.

- M points in 2D space, distributed across N chares in a **chare array**
- each Lloyd iteration is one **broadcast** + two **reductions** over a chare array.
- Charm4Py **futures** are used as reduction targets



Algorithmic details

1. `main()` generates K random points in $[0, 1)^2$ as initial centroid guesses.
2. **TODO: `main()` broadcasts the K centroids to every Points chare via the `assign()` entry method.**
3. Each chare assigns each of its local points to the nearest centroid (by squared Euclidean distance)
4. **TODO: each chare then contributes to two sum reductions:**
 - `counts` -- length- K integer array, number of local points assigned to each cluster,
 - `coords` -- length- $2K$ double array, sum of x and y coordinates of local points assigned to each cluster (interleaved: $x_0, y_0, x_1, y_1, \dots$).
5. **TODO: `main()` receives both reduced arrays via futures.**
6. Centroids are updated.
7. Convergence: when no centroid coordinate moves by more than THRESHOLD (0.001) between successive iterations, stop. Otherwise, repeat.

